

Tail Design & Performance Analysis

Week 8 Report | Group 11

UAV Design Course

April 9, 2026

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- 2 Vertical Tail Design
- 3 CG Estimation & Stability Verification
- 4 Parasitic Drag & Drag Polar
- 5 Performance Analysis
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Need for Horizontal Tail Contribution

Wing–Body Pitching Behaviour

- Overall $C_M < 0 \Rightarrow$ **net nose-down** tendency
- $C_{M_\alpha} > 0$ for wing–body: **longitudinally unstable**
- Any increase in α drives further nose-up moment

Solution

Horizontal tail must produce a **nose-up** (positive) pitching moment by generating **downward lift**, counteracting wing–body instability.

Tail Pitching Moment Equations

$$C_{m_t} = -\eta_t V_H C_{L_t}$$
$$C_{m_{\alpha,t}} = -\eta_t V_H C_{L_{\alpha,t}} \left(1 - \frac{d\epsilon}{d\alpha}\right); \eta_t = 0.99$$

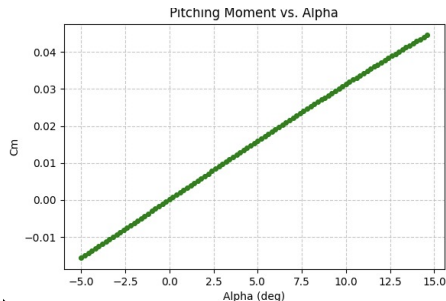


Figure: C_m v. α for fuselage

Horizontal Tail Sizing — Key Parameters

Design Targets

- Static Margin: $K_n = 10\% \bar{c}$
- $S_t = 25\%$ of wing area (from comparable UAVs)
- $AR_{HT} = 4.5$
- Neutral Point (stick-fixed): $X_{NP} = 0.2676 \text{ m}$

Tail Arm — Governing Equation

$$l_t = \frac{\left[(X_{NP} - X_{ac,w}) + \frac{C_{m,f}}{C_{L_\alpha}} \right] S}{\eta S_t \frac{C_{L_{\alpha,t}}}{C_{L_\alpha}} \left(1 - \frac{d\epsilon}{d\alpha} \right)} = 0.7 \text{ m}$$

Table: Aerodynamic & Geometric Parameters

Parameter	Value
α_{trim}	3°
X_{cg}	0.24106 m
$X_{ac,w}$	0.20193 m
K_n	$0.10 \bar{c}$
S_t	0.1327 m^2
η	0.99
$C_{L_{\alpha,t}}$	3.831 rad^{-1}
C_{L_α}	4.570 rad^{-1}
$C_{m_{ac,w}}$	-0.0683
AR_{HT}	4.5

Tail Setting Angle & Aerofoil Selection

Trim Condition — Setting Angle

Total pitching moment about CG:

$$C_{m_{cg}} = C_{m_{ac,w}} + C_{L_w}(X_{cg} - X_{ac,w}) - \eta \left(\frac{S_t}{S} \right) C_{L_{\alpha,t}} (\alpha + i_t - \frac{d\epsilon}{d\alpha} \alpha) \frac{l_t}{\bar{c}} + C_{m_f}$$

Setting $C_{m_{cg}} = 0 \Rightarrow i_t = 0.1757^\circ$

Why NACA 0012?

- **Symmetric:** $C_{m_{ac}} = 0$ — no built-in lift bias
- Works for both +ve and -ve lift requirements
- Simplifies longitudinal stability analysis
- Adjustable via incidence or elevator deflection

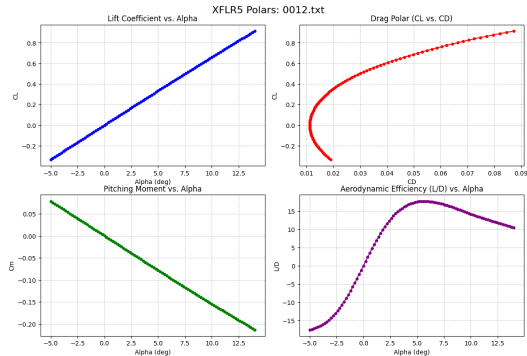


Figure: Aerodynamic forces and moments for NACA 0012 Tail

$$C_{L_{\alpha,t}} = 3.831 \text{ rad}^{-1}$$

Vertical Tail Design — Directional Stability

Weathercock Stability Requirement

$$C_{n_\beta} > 0 \iff C_{n_{\beta v}} = \eta_v \frac{S_v l_v}{S b} C_{L_{\alpha v}} > 0$$
$$C_{n_\beta} = C_{n_{\beta w}} + C_{n_{\beta f}} + C_{n_{\beta v}}$$

Sizing Approach

- Common small-UAV design target:
 $S_v/S = 0.09$
- $S_v = 0.09 \times 0.5307 = 0.0478 \text{ m}^2$
- $AR_v = 1.5$, rectangular (no taper)
- $b_v = \sqrt{AR_v \cdot S_v} = 0.268 \text{ m}$
- $c_v = S_v/b_v = 0.178 \text{ m}$
- $l_v = l_t = 0.7814 \text{ m}$ (same fuselage station)
- Aerofoil: **NACA 0012**

Why T-Tail Configuration?

- Horizontal tail exits wing wake $\Rightarrow$ higher η_t
- Reduced wing-fuselage aerodynamic interference
- Vertical tail gives support to horizontal tail

Table: Vertical Tail Geometry

Parameter	Value
Area S_v	0.0478 m^2
Arm l_v	0.7814 m
AR_v	1.5
Span b_v	0.268 m
MAC_v	0.178 m
Setting Angle	0°

CG Estimation with Full Empennage

Mass Distribution (SolidWorks)

Component	Mass (kg)
Fuselage + Tail Boom	1.146
Wing + Aileron	1.399
Motors ($\times 2$)	0.264
Propeller	0.096
Battery	1.524
Landing Gear	0.565
Drop Payload	1.200
Tail	0.296
Total	6.484

Computed CG Location (from nose)

- Longitudinal: **254 mm**

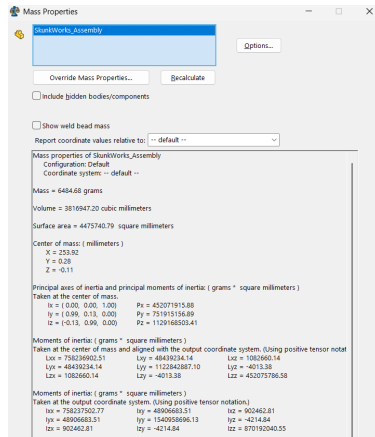


Figure: Mass Properties of CAD from Solidworks

Longitudinal Stability Verification

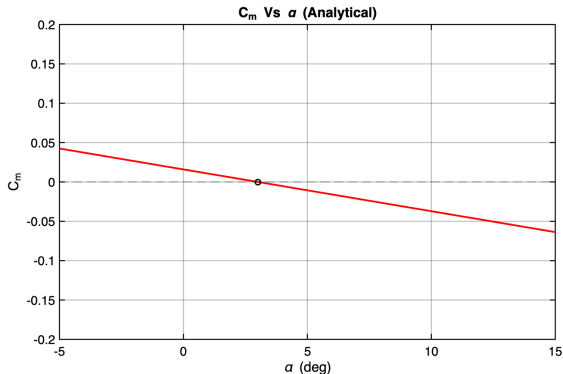


Figure: C_m v. α for UAV

Computed Stability Derivatives

Param.	w/o Tail	with Tail
$d\epsilon/d\alpha$	0.5051 rad ⁻¹	
$C_{m_{\alpha f}}$	0.1797 rad ⁻¹	
$C_{m_{\alpha}}$	+0.384	-0.304
C_{m_0}	0.0201	0.0159
C_{L_t}	-0.0251	+0.1110

Trim: $C_{m_{cg}} = 0$ at $\alpha_{\text{trim}} = 3^\circ$

Stability Conditions

$$\underbrace{C_{m_{\alpha}} < 0}_{\text{stable}} \quad \text{and} \quad \underbrace{C_{m_0} > 0}_{\text{trimable}}$$

$i_w = 3^\circ$ for $\alpha_{FRL} = 0$ at cruise.

Empennage Design Summary & CAD Model

Final Empennage Parameters

Parameter	Value
<i>Horizontal Tail</i>	
Area S_t	0.1327 m ²
Arm l_t	0.7814 m
AR_{HT}	4.5
Span b_t	0.773 m
MAC $\bar{c}_t$	0.172 m
Setting Angle i_t	0.1757°
<i>Vertical Tail</i>	
Area S_v	0.0478 m ²
Arm l_v	0.7814 m
AR_{VT}	1.5
Span b_v	0.268 m
MAC $\bar{c}_v$	0.178 m
Setting Angle i_v	0°

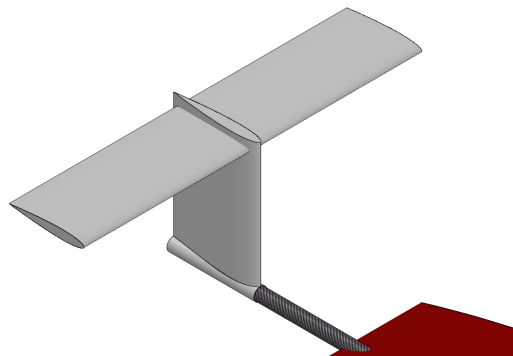


Figure: Zoomed View of the Tail Section

Parasitic Drag Estimation

Component Buildup Method

$$(C_{D_0})_{\text{aircraft}} = \frac{\sum C_{f_c} F F_c Q_c S_{\text{wet},c}}{S_{\text{ref}}}$$

At altitude 200 m: $\rho = 1.22 \text{ kg/m}^3$, $V = 18.8 \text{ m/s}$,
 $M \approx 0.053$ (fully incompressible).

Key Modelling Decisions

- **Compressibility term omitted** for form factor (avoids $FF < 1$ at low M)
- **Fuselage:** treated as lifting body (NACA 0025, $t/c = 0.25$) not a cylinder
- **Square-sided penalty** $P_{sq} = 0.2$ applied for flat fuselage sides: $FF_{\text{fus}} = FF_{\text{base}} + (FF_{\text{base}} - 1)P_{sq}$

Component Drag Breakdown (Table 6.2)

Comp.	C_f	FF	Q	S_{wet}	$C_f \cdot FF \cdot Q \cdot S_w$
Wing	0.00541	1.210	1.00	0.93	0.00609
H. Tail	0.00605	1.261	1.00	0.22	0.00168
V. Tail	0.00587	1.261	1.05	0.09	0.00070
Fuselage	0.00437	2.069	1.00	0.75	0.00678
Total C_{D_0}					0.02873

Key Aerodynamic Results

$$C_{D_0} = 0.02873 \quad \left(\frac{L}{D} \right)_{\text{max}} = 12.55$$

Drag Polar

Drag Polar Equation

$$C_D = C_{D_0} + \frac{C_L^2}{\pi A R e}$$

$$C_{D_0} = 0.02873, \quad A R = 7.53, \quad e = 0.76$$

Maximum L/D

$$\left(\frac{L}{D}\right)_{\max} = \frac{1}{2} \sqrt{\frac{\pi A R e}{C_{D_0}}} = \mathbf{12.55}$$

Significance

- Determines optimal cruise & loiter speeds
- Sets maximum range & endurance conditions
- Guides propulsion power budgeting

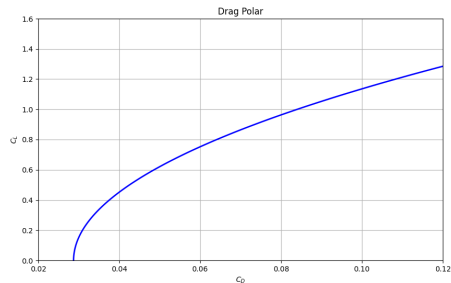


Figure: Drag Polar

Maximum Range & Endurance

Operating Conditions

- **Max Range:** $(C_L/C_D)_{\max} \Rightarrow C_{D0} = C_{Di}$
- **Max Endurance:** $(C_L^{3/2}/C_D)_{\max} \Rightarrow 3C_{D0} = C_{Di}$

Results

$$R_{\max} = 83.15 \text{ km}$$

$$E_{\max} = 2 \text{ hr } 5 \text{ min}$$

Design range requirement ✓

Battery-Electric Formulae

$$R_{\max} = \eta_p \eta_m \eta_b (E_{\text{bat}}) \frac{V_{R_{\max}}}{P_{R_{\max}}}$$

$$E_{\max} = \frac{\eta_p \eta_m \eta_b (E_{\text{bat}})}{P_{E_{\max}}}$$

$$E_{\text{bat}} = 959\,040 \text{ J}; \quad \eta_p = 0.7, \quad \eta_m = 0.9, \quad \eta_b = 0.9$$

Recomputed Optimum Speeds

Segment	V (m/s)	T (N)	P (W)
Climb-1	22.67	11.40	258.5
Cruise-1	18.78	5.08	95.4
Loiter-1	22.72	5.87	133.3
Climb-2	20.47	9.30	190.4
Cruise-2	16.96	4.14	70.3
Loiter-2	20.51	4.78	98.1

Power Required & Rate of Climb

Power Required

$$P_{\text{req}} = \frac{1}{2} \rho V^3 S_{\text{ref}} \left(C_{D0} + \frac{1}{\pi A R e} \left(\frac{2W}{\rho V^2 S_{\text{ref}}} \right)^2 \right)$$

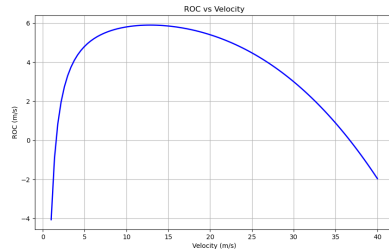
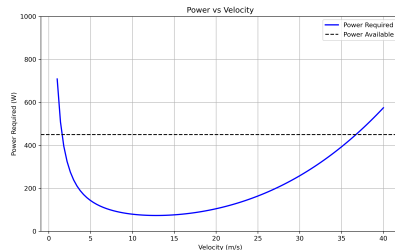
Rate of Climb

$$ROC = \frac{P_{\text{avl}} - P_{\text{req}}}{W}$$

Peak ROC occurs near $V \approx 10 \text{ m/s}$

Observations

- Cruise at 18.8 m/s is well within the available power band
- Excess power margin confirmed across mission segments



Flight Envelope (V-n Diagram)

Characteristic Speeds

Speed	Sym.	Before	After
Stall Speed	V_s	12.40	11.21
Max Speed	$V_{\max}$	24.00	24.00
Design Dive	V_d	30.00	30.00
Manoeuvre	V_a	23.19	20.97
Neg. Stall	V_g	19.35	17.50

Load Factor Relations

$$n = \frac{\rho V^2 S C_L}{2W}, \quad V_a = V_s \sqrt{n_{\max}}$$
$$V_d = 1.25 V_{\max}$$

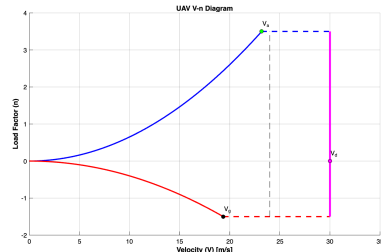


Figure: V-n diagram before payload drop

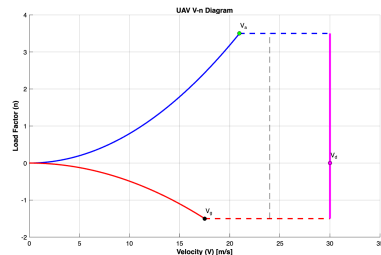


Figure: V-n diagram after payload drop

Stability & Tail Design

- Wing-body configuration: **unstable**
- T-tail (NACA 0012) added $\Rightarrow C_{m_\alpha} < 0$, $C_{m_0} > 0$
- $i_t = 0.1757^\circ$, $i_w = 3^\circ$
- Horizontal tail: 0.1327 m^2 , arm 0.78 m
- Vertical tail: 0.0478 m^2 , arm 0.78 m
- Static Margin: **8%** — stable

Performance

- $C_{D_0} = 0.02873$
- $(L/D)_{\max} = 12.55$
- $R_{\max} = \mathbf{83.15 \text{ km}}$
- $E_{\max} = \mathbf{2 \text{ hr } 5 \text{ min}}$
- Cruise power: $\sim 95 \text{ W}$
- V-n envelopes computed for both payload states

UAV is aerodynamically trimmed, statically stable, and meets mission performance targets.

Next Steps: Stability Analysis $\rightarrow$ Control system design

Thank You

Questions & Discussion

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